

Modeling Process Reflection

Accomplishments

Technical Achievements

The feature engineering pipeline produced measurable value. Seven engineered features were created from domain knowledge and EDA findings, with *medication_change_count* replacing 21 near-zero-variance medication columns (reducing one-hot encoded features from ~92 to ~6 for that group alone) and *total_visits_prior_year* compositing three separate utilization columns into a single index that ranked in the SHAP top 10. The ICD-9 grouping function, following Strack et al.'s published methodology, collapsed ~700 unique diagnosis codes per column into 9 clinically validated categories -- a 98% cardinality reduction that made the diagnosis features tractable for tree-based models.

The two-stage hyperparameter tuning approach (RandomizedSearchCV for exploration, then GridSearchCV for refinement) was the right call for this problem. XGBoost's parameter space had 7,680 combinations; a full grid search would have taken ~34 hours while RandomizedSearch sampled 80 combinations in under 2 minutes and found the optimal region. The key tuning insight was that reducing the learning rate from 0.1 to 0.03 and the depth from 6 to 4 -- slower, more careful boosting with less overfitting -- produced the largest AUC improvement of any model (+0.024).

The three-way class imbalance comparison (class_weight='balanced', SMOTE, and no handling) was the most impactful experiment in Phase 4. The no-handling condition achieved a recall of 0.008 -- it caught less than 1% of readmissions by defaulting to the majority class prediction. That single number is the strongest argument for why imbalance handling is non-negotiable in clinical classification.

Process Achievements

The project followed CRISP-DM rigorously, with each phase building on validated outputs from the prior phase. The test set was locked from Phase 3 through Phase 5 and used exactly once in Phase 6 -- there was no implicit leakage from threshold tuning on the test set or feature selection informed by test performance. The utility functions defined in Phase 4 (evaluate_model, predict_at_threshold, compute_business_impact) were reused across Phases 4-6, following peer review feedback to reduce code duplication. This made the metric reporting consistent across all evaluation contexts.

The notebook's narrative structure -- where every chart cell is followed by an interpretation cell citing actual results -- ensures that a reviewer can read the markdown cells alone and understand the full story without executing any code. The EDA hypotheses stated in Section 2.10 were explicitly revisited in Section 6.8 with SHAP evidence, closing the CRISP-DM loop between data understanding and evaluation.

Opportunity for Growth

The precision-recall tradeoff is the biggest limitation I'd address before a real deployment. At the default threshold (0.50), the model achieves 62% recall but only 12% precision -- meaning 88% of flagged patients would not actually be readmitted. The recommended operating threshold (0.30) catches 92% of readmissions but flags 75% of all patients. In a resource-constrained hospital, that volume of interventions may not be feasible. The business impact simulation shows it's still net-positive financially, but the operational burden on case management teams is a real concern that the model alone doesn't solve.

The has_A1c_measured feature was a disappointment. It showed a clear univariate association with readmission in Phase 2 and directly replicated the Strack et al. (2014) finding, but it did not appear in the SHAP top 20. Its signal was likely absorbed by correlated features (diabetesMed, number_diagnoses) in the tree ensemble. This is an honest result, but I could have explored interaction terms or a dedicated A1c sub-model to preserve that clinical signal.

Probability calibration is an unaddressed gap. The model produces well-ranked probabilities (good AUC) but may not be well-calibrated -- a predicted 0.30 probability may not correspond to a true 30% readmission rate. For clinical communication, calibrated probabilities matter because clinicians interpret them as actual risk. Platt scaling or isotonic regression would address this, and I'd add a reliability diagram in a future iteration.

Finally, the dataset is from 1999-2008 and uses ICD-9 codes. US hospitals transitioned to ICD-10 in 2015. The 9-category clinical grouping would need remapping, and the specific model coefficients may not transfer to current practice. This is acknowledged in the limitations section but is worth emphasizing -- this is a methodology demonstration, not a deployment-ready model.

Feedback Request

I'd appreciate specific feedback on the threshold optimization and business impact framing.

The model's optimal F1 threshold (0.50) and the recommended business operating threshold (0.30) are different. At 0.50, F1 is maximized but recall is 62%. At 0.30, recall is 92% but F1 drops and the number of flagged patients nearly doubles. I chose to recommend the 0.30 threshold based on the cost asymmetry: a missed readmission costs \$13,000 in penalties while a false alarm costs \$500 in intervention. But I'm unsure whether the presentation should lead with the F1-optimal threshold (which is more standard in ML) or the cost-optimal threshold (which is more relevant to the business stakeholder).

Specifically: When you look at Section 6.4 (Threshold Optimization) and Section 6.7 (Business Impact Simulation), does the transition from statistical metrics to dollar-denominated recommendations feel logical, or does it need a stronger bridge? Is the 26:1 cost ratio (\$13K penalty vs. \$500 intervention) a convincing enough justification for operating at a threshold that flags 75% of all patients?